

MI-18: a counterexample to q -permanent monotonicity

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The universal monotonicity assertion in MI-18 is false. An ordered family of 144 vectors in $\mathbb{C}^2$ gives a positive semidefinite Gram matrix whose q -permanent has a strictly negative derivative at $q = 1$. Adding a sufficiently small positive multiple of the identity gives a positive definite counterexample as well. The proof reduces the sign of that derivative to a finite calculation with Gaussian integers; all input integers are printed in the appendix.

The certificate and Lean formalization were supplied by Kenta Kitamura in [issue #329](#). The source used here is fixed at [commit 4200da4fc1a132d69c23fb877795b5b72089544b](#). Codex prepared this conventional mathematical exposition and the accompanying independent Python checker. This is AI-prepared prose, not a manuscript written or approved by Kitamura. The source repository discloses AI assistance in its formalization and certificate development; no claim of external human review or discovery priority is made here.

1. Statement and construction

We index rows and columns by $0, \dots, n-1$. This simply shifts the indices in MI-18 by one and preserves their order. For $\sigma \in S_n$, let

$$\text{inv}(\sigma) = \#\{(i, j) : 0 \leq i < j < n, \sigma(i) > \sigma(j)\}.$$

For a complex matrix $A = (A_{ij})$, put

$$P_q(A) = \sum_{\sigma \in S_n} q^{\text{inv}(\sigma)} \prod_{i=0}^{n-1} A_{i, \sigma(i)}, \quad 0^0 = 1. \quad (1)$$

In particular, $P_1(A) = \text{per}(A)$. If A is Hermitian, (1) is real for real q : conjugating a summand replaces σ by σ^{-1} , and $\text{inv}(\sigma^{-1}) = \text{inv}(\sigma)$. For the latter equality, an inversion $i < j$, $\sigma(i) > \sigma(j)$ corresponds to the inversion $(\sigma(j), \sigma(i))$ of σ^{-1} . Thus (1) defines an ordinary real polynomial in this case.

Let the ordered triples (a_j, u_j, v_j) be those in the appendix, and set

$$n = 144, \quad b_j = u_j + \mathbf{i}v_j, \quad V_{j,:} = (a_j, b_j), \quad H = VV^*. \quad (2)$$

Theorem. The matrix H is Hermitian positive semidefinite, has positive diagonal, is non-diagonal, and satisfies

$$\left. \frac{d}{dq} P_q(H) \right|_{q=1} < 0. \quad (3)$$

Moreover, there exist $\varepsilon > 0$ and $0 < q_1 < q_2 = 1$ for which

$$A = H + \varepsilon I_{144} > 0, \quad P_{q_2}(A) < P_{q_1}(A). \quad (4)$$

Consequently both the positive semidefinite version stated in MI-18 and the positive definite version of the Bapat-Lal conjecture fail. The construction gives order 144; it makes no assertion that this is the smallest possible order. The proof of (4) is existential in ε and q_1 .

2. An endpoint derivative identity

For an arbitrary complex $n \times n$ matrix with $n \geq 2$, write

$$D(A) = \left. \frac{d}{dq} P_q(A) \right|_{q=1} = \sum_{\sigma \in S_n} \text{inv}(\sigma) \prod_i A_{i, \sigma(i)}.$$

If $I = \{i, j\}$ and $J = \{k, \ell\}$ with $i < j$ and $k < \ell$, write $A[I, J]$ for the corresponding 2×2 submatrix, with these increasing orders. The complementary submatrix is $A[I^c, J^c]$. The permanent of the empty matrix is understood to be 1. Define

$$T(A) = \sum_{\substack{I, J \subseteq \{0, \dots, n-1\} \\ |I|=|J|=2}} \det A[I, J] \text{ per } A[I^c, J^c].$$

Lemma 1. For every such matrix,

$$2D(A) = \binom{n}{2} \text{per}(A) - T(A). \quad (5)$$

Proof. Fix $i < j$ and put $M_\sigma = \prod_r A_{r, \sigma(r)}$. Separate the permanent into the sums E_{ij} over permutations satisfying $\sigma(i) < \sigma(j)$ and O_{ij} over those satisfying $\sigma(i) > \sigma(j)$. These two alternatives exhaust all permutations, so

$$E_{ij} + O_{ij} = \text{per}(A).$$

For each permutation in E_{ij} , put $k = \sigma(i) < \ell = \sigma(j)$ and pair it with the permutation that exchanges the images of i and j . Their difference is

$$M_\sigma - M_{\sigma \circ (i \ j)} = (A_{ik}A_{j\ell} - A_{i\ell}A_{jk}) \prod_{r \notin \{i, j\}} A_{r, \sigma(r)}.$$

Fixing $k < \ell$ and summing over all bijections between the remaining row and column sets gives exactly

$$E_{ij} - O_{ij} = \sum_{k < \ell} \det A[\{i, j\}, \{k, \ell\}] \text{per } A[\{i, j\}^c, \{k, \ell\}^c].$$

Now sum over $i < j$. A given M_σ occurs in $\sum_{i < j} O_{ij}$ exactly $\text{inv}(\sigma)$ times. Hence

$$T(A) = \sum_{i < j} (E_{ij} - O_{ij}) = \binom{n}{2} \text{per}(A) - 2D(A),$$

as required. No positivity assumption was used.

3. Permanents of two-coordinate Gram matrices

For polynomials of degree at most m , define the weighted coefficient pairing

$$\langle f, h \rangle_m = \sum_{k=0}^m k!(m-k)! [z^k]f \overline{[z^k]h}, \quad \|f\|_m^2 = \langle f, f \rangle_m. \quad (6)$$

Consider two families of m vectors (α_i, β_i) and (γ_j, δ_j) , and the matrix

$$B_{ij} = \alpha_i \overline{\gamma_j} + \beta_i \overline{\delta_j}.$$

Set $p_X(z) = \prod_i (\alpha_i + \beta_i z)$ and $p_Y(z) = \prod_j (\gamma_j + \delta_j z)$.

Lemma 2. One has

$$\text{per}(B) = \langle p_X, p_Y \rangle_m. \quad (7)$$

Proof. In the expansion of a permanent summand, let R be the rows in which the second coordinate is chosen, and let C be the columns matched to these rows. If $|R| = |C| = k$, exactly $k!(m-k)!$ permutations map R bijectively onto C and R^c onto C^c . Their chosen-coordinate products are identical. Summing over R gives

$$\sum_{|R|=k} \prod_{i \in R} \beta_i \prod_{i \notin R} \alpha_i = [z^k] p_X,$$

and the column sum is the conjugate of $[z^k] p_Y$. Summing over k proves (7), including $m = 0$ by the empty-product convention.

Return to arbitrary ordered rows (a_j, b_j) of a two-column matrix V , and put

$$\ell_j(z) = a_j + b_j z, \quad p(z) = \prod_{j=0}^{n-1} \ell_j(z), \quad H = VV^*.$$

Taking the two families equal in (7) yields

$$\text{per}(H) = \|p\|_n^2. \quad (8)$$

For $I = \{i, j\}$, $i < j$, write

$$w_I = a_i b_j - a_j b_i, \quad p_I(z) = \prod_{r \notin I} \ell_r(z), \quad F(z) = \sum_{|I|=2} w_I p_I(z).$$

The 2×2 submatrix $H[I, J]$ equals $V[I, :]V[J, :]^*$, so

$$\det H[I, J] = w_I \overline{w_J}.$$

The complementary submatrix is likewise a mixed Gram matrix. Applying (7) with $m = n - 2$ gives

$$\text{per } H[I^c, J^c] = \langle p_I, p_J \rangle_{n-2}.$$

Substitution into $T(H)$ and expansion of the pairing now give

$$\begin{aligned} T(H) &= \sum_{I,J} w_I \overline{w_J} \langle p_I, p_J \rangle_{n-2} \\ &= \sum_{k=0}^{n-2} k! (n-2-k)! \left| \sum_I w_I [z^k] p_I \right|^2 = \|F\|_{n-2}^2. \end{aligned} \quad (9)$$

Together, (5), (8), and (9) prove the rank-two endpoint formula

$$2D(H) = \binom{n}{2} \|p\|_n^2 - \|F\|_{n-2}^2. \quad (10)$$

4. The wedge polynomial and its integer recurrence

To compute F without summing over all pairs, define

$$g(z) = \sum_{j=0}^{n-1} (n-1-2j) b_j \prod_{r \neq j} \ell_r(z).$$

Lemma 3. $F = -g$. In particular, g has degree at most $n-2$.

Proof. The identity

$$a_i b_j - a_j b_i = \ell_i(z) b_j - \ell_j(z) b_i$$

implies

$$w_{\{i,j\}} p_{\{i,j\}}(z) = b_j \prod_{r \neq j} \ell_r(z) - b_i \prod_{r \neq i} \ell_r(z).$$

In the sum over $i < j$, the term $b_j \prod_{r \neq j} \ell_r$ receives j positive contributions from indices below j , and $n-1-j$ negative contributions from indices above it. Its net coefficient is therefore $2j-n+1$. This proves $F = -g$. Each summand defining F has degree at most $n-2$, proving the degree assertion without dividing by any ℓ_j or coordinate.

For the certificate (2), put

$$P = \|p\|_{144}^2, \quad S = \|g\|_{142}^2.$$

Then (10) becomes

$$2D(H) = 10296P - S, \quad \binom{144}{2} = 10296. \quad (11)$$

Here is a complete finite procedure for computing the two integers in (11). Use ascending coefficients and start with $p^{(0)} = 1$ and $g^{(0)} = 0$. For $j = 0, \dots, 143$, successively form

$$\begin{aligned} p^{(j+1)}(z) &= (a_j + b_j z) p^{(j)}(z), \\ g^{(j+1)}(z) &= (a_j + b_j z) g^{(j)}(z) + (143 - 2j) b_j p^{(j)}(z). \end{aligned} \quad (12)$$

Induction shows that $p^{(j)}$ is the product of the first j factors, and $g^{(j)}$ is the sum with one marked factor among those first j factors, using the fixed final weights $143-2r$. Thus $p^{(144)} = p$

and $g^{(144)} = g$. More explicitly, if $c_{j,k} = [z^k]p^{(j)}$ and $d_{j,k} = [z^k]g^{(j)}$, take $c_{0,0} = 1$, all other initial coefficients zero, and use

$$\begin{aligned} c_{j+1,k} &= a_j c_{j,k} + b_j c_{j,k-1}, \\ d_{j+1,k} &= a_j d_{j,k} + b_j d_{j,k-1} + (143 - 2j)b_j c_{j,k}. \end{aligned}$$

Coefficients outside their ranges are zero. Every operation is addition or multiplication of Gaussian integers. With (x, y) representing $x + iy$, these operations are

$$(x, y) + (u, v) = (x + u, y + v), \quad (x, y)(u, v) = (xu - yv, xv + yu).$$

After all 144 steps, the coefficient $d_{144,143}$ is exactly zero, as Lemma 3 also predicts. Take

$$\begin{aligned} P &= \sum_{k=0}^{144} k!(144 - k)!((\operatorname{Re} c_{144,k})^2 + (\operatorname{Im} c_{144,k})^2), \\ S &= \sum_{k=0}^{142} k!(142 - k)!((\operatorname{Re} d_{144,k})^2 + (\operatorname{Im} d_{144,k})^2). \end{aligned}$$

Evaluation on the appendix gives the exact integer inequalities

$$P > 0, \quad 10300P < S < 10301P. \quad (13)$$

The positivity of P also follows directly: all the a_j are nonzero, so $[z^0]p = \prod_j a_j \neq 0$. The integers P and S have 768 and 772 decimal digits, respectively; their full decimal expansions are unnecessary for the sign test. For orientation only, division gives

$$\frac{S}{P} \approx 10300.4591950231122, \quad \frac{D(H)}{P} \approx -2.2295975115561.$$

These decimal approximations are not used to establish (13). The exact lower bound alone gives

$$2D(H) = 10296P - S < -4P < 0,$$

which proves (3).

The machine-readable [certificate](#) contains only the ordered triples and provenance meta-data. The separately written [Python checker](#) uses arbitrary-precision integers from the Python standard library. It checks the coefficient cancellation, (13), the nonzero off-diagonal entry used below, and an additional recurrence for F . In this second recurrence, $p^{(j)}$ has the meaning above and $F^{(j)}$ is the wedge sum for the first j rows. Adjoining row j gives

$$F^{(j+1)} = \ell_j F^{(j)} + j b_j p^{(j)} - \ell_j \frac{d}{dz} p^{(j)}, \quad F^{(0)} = 0.$$

Indeed the new pairs contribute $b_j \sum_{r < j} a_r \prod_{s < j, s \neq r} \ell_s - a_j (p^{(j)})'$, and the first sum equals $j p^{(j)} - z(p^{(j)})'$. This yields the displayed recurrence. The checker verifies that its final coefficient list is exactly the negative of that produced for g by (12).

From the repository root, the complete numerical reproduction command is

```
python3 matrix-inequalities-and-norms/MI-18/verify_certificate.py
```

It returns PASS together with the exact rational bounds and hashes of the two computed integers. The finite arithmetic is independently reproducible from the appendix and (12); no copy of the external Lean project is needed for this calculation.

5. Positive definiteness and strict decrease

For every vector $x \in \mathbb{C}^{144}$,

$$x^* H x = x^* V V^* x = \|V^* x\|_2^2 \geq 0,$$

so H is Hermitian positive semidefinite. Its diagonal entries are $a_j^2 + u_j^2 + v_j^2 > 0$. The first two rows give

$$\begin{aligned} H_{0,1} &= (-60)(-55) + (72 - 35i)(80 + 21i) \\ &= 9795 - 1288i \neq 0, \end{aligned}$$

so H is non-diagonal. This verifies every matrix hypothesis in the positive semidefinite formulation of MI-18.

Now put $A_\varepsilon = H + \varepsilon I_{144}$. For every $\varepsilon > 0$ and $x \neq 0$,

$$x^* A_\varepsilon x = \|V^* x\|_2^2 + \varepsilon \|x\|_2^2 > 0.$$

The off-diagonal entry computed above is unaffected by this perturbation. Also $D(A_\varepsilon)$ is a polynomial in ε , by the finite sum defining D . It is therefore continuous at zero. Since $D(A_0) = D(H) < 0$, there exists $\varepsilon_0 > 0$ such that

$$D(A_\varepsilon) < 0 \quad (0 < \varepsilon < \varepsilon_0).$$

Choose any such ε , and let $f(q) = P_q(A_\varepsilon)$. The function f is a real polynomial with $f'(1) < 0$. By the definition of the derivative, for all sufficiently small $h > 0$,

$$\frac{f(1) - f(1-h)}{h} < 0.$$

Taking $h < 1$ gives $q_1 = 1 - h \in (0, 1)$, $q_2 = 1$, and $f(q_2) < f(q_1)$. This proves (4). The same argument applied directly to H gives a strict decrease for the positive semidefinite witness.

The proof specifies neither a numerical perturbation size nor a numerical comparison point. In particular, it does not assert a decrease for a separately proposed fixed choice such as $10^{70}H + I$ at $q = 1 - 10^{-70}$. It also gives no counterexample restricted to real symmetric matrices. These additional statements are unnecessary to disprove the universal complex Hermitian conjecture.

Appendix. The ordered integer certificate

The table specifies (a_j, u_j, v_j) with $b_j = u_j + iv_j$. Each block lists a different range of row indices; use the displayed index j to reconstruct the order $0, 1, \dots, 143$. Simultaneously permuting rows and columns need not preserve the inversion-weighted polynomial, so this order is part of the mathematical data.

The triples are transcribed from Kitamura's [immutable certificate source](#). The accompanying JSON repeats precisely this small dataset, not the external proof repository. The source repository is distributed under [Apache-2.0](#); the data attribution and license provenance are recorded in the JSON.

j	(a_j, u_j, v_j)	j	(a_j, u_j, v_j)	j	(a_j, u_j, v_j)
0	(−60, 72, −35)	48	(61, 58, 54)	96	(69, −10, −71)
1	(−55, 80, −21)	49	(57, 72, 40)	97	(95, 30, −1)
2	(52, −85, −1)	50	(42, 88, −22)	98	(78, 7, −62)
3	(49, −84, 21)	51	(−53, −83, −15)	99	(84, 17, −51)
4	(63, −61, 48)	52	(38, 59, −71)	100	(90, 26, −36)
5	(52, −68, 52)	53	(42, 76, −50)	101	(95, 27, −18)
6	(68, −51, 53)	54	(40, −7, −91)	102	(92, −7, 38)
7	(39, −83, 40)	55	(41, 34, −84)	103	(98, 19, 6)
8	(−41, 91, 5)	56	(79, 5, 61)	104	(66, −37, −65)
9	(45, −58, 68)	57	(77, 29, 57)	105	(94, 16, −29)
10	(59, −46, 66)	58	(74, 46, 49)	106	(98, 17, −12)
11	(30, −95, 7)	59	(68, 65, 33)	107	(90, 8, −42)
12	(32, −60, 73)	60	(−55, −79, 27)	108	(99, 7, 14)
13	(48, −83, −27)	61	(65, 75, 12)	109	(97, −1, 25)
14	(49, −32, 81)	62	(61, 79, −11)	110	(−77, 20, 61)
15	(−21, 79, −57)	63	(52, −56, −64)	111	(100, 6, −1)
16	(−44, 80, 41)	64	(50, −29, −82)	112	(−85, 6, 52)
17	(63, −29, 72)	65	(−55, −65, 53)	113	(87, −27, 41)
18	(36, −19, 91)	66	(77, 54, 33)	114	(66, −54, −53)
19	(21, −23, 95)	67	(−51, −4, 86)	115	(92, −19, 33)
20	(14, −99, −4)	68	(75, 65, 15)	116	(99, 1, −15)
21	(−33, 80, 51)	69	(−54, −38, 75)	117	(96, −2, −28)
22	(−73, 36, −59)	70	(84, −11, 54)	118	(92, −9, −39)
23	(−48, 1, −88)	71	(71, 69, −10)	119	(84, −22, −49)
24	(−19, 75, 64)	72	(61, 49, −62)	120	(−98, 12, −14)
25	(31, 21, 93)	73	(66, 63, −41)	121	(76, −39, −52)
26	(59, −6, 80)	74	(84, 37, 40)	122	(99, −12, 3)
27	(2, 93, −37)	75	(86, 18, 48)	123	(99, −13, −9)
28	(5, 84, −55)	76	(−72, −62, 31)	124	(96, −17, −24)
29	(16, 70, 70)	77	(60, 14, −79)	125	(95, −23, 20)
30	(45, 28, 85)	78	(81, 59, −3)	126	(91, −27, −33)
31	(−24, −75, −61)	79	(84, 53, 13)	127	(85, −36, −39)
32	(27, −43, −86)	80	(79, 57, −20)	128	(95, −27, −16)
33	(34, 65, 68)	81	(88, 41, 25)	129	(69, −61, −39)
34	(70, −7, 72)	82	(90, 26, 36)	130	(96, −28, −2)
35	(−61, −18, −77)	83	(73, 45, −51)	131	(86, −40, 32)
36	(45, 60, 67)	84	(69, 30, −66)	132	(78, −51, −36)
37	(54, 42, 73)	85	(89, 4, 46)	133	(94, −34, 8)
38	(20, 47, −86)	86	(91, 39, 13)	134	(90, −39, 22)
39	(28, 96, 6)	87	(89, 45, −2)	135	(89, −42, −17)
40	(76, −20, 62)	88	(61, −22, −76)	136	(83, −52, −22)
41	(25, 86, −44)	89	(82, 42, −39)	137	(88, −47, −7)
42	(28, 10, −96)	90	(86, 43, −27)	138	(−73, 64, 24)
43	(41, −51, −76)	91	(90, 41, −15)	139	(87, −49, 7)
44	(44, 82, 37)	92	(71, 9, −70)	140	(84, −50, 22)
45	(71, 16, 68)	93	(80, 29, −53)	141	(76, −63, −12)
46	(66, 39, 64)	94	(95, 24, 20)	142	(82, −57, 10)
47	(42, 91, 6)	95	(94, 12, 31)	143	(79, −62, −1)

Sources and scope

- R. B. Bapat and A. K. Lal, *Inequalities for the q -permanent*, Linear Algebra and its Applications 197-198 (1994), 397-409, [original paper](#).
- L. Mitchell, *A note on Bapat's q -permanent conjecture*, Operators and Matrices 14 (2020), 915-919, [primary paper](#), for the positive semidefinite formulation and earlier special cases.
- Kenta Kitamura, [submission in issue #329](#), supplying the certificate and formalization used here.
- Kenta Kitamura, [complete proof entry point](#) and [source README](#), at the fixed commit cited above. The formal target there is the positive definite complex Hermitian conjecture on $[-1, 1]$.
- [MI-18's canonical statement](#). Its broader positive semidefinite formulation is contradicted directly by H , as shown in Sections 4 and 5.

The mathematical derivation above and the integer checker establish the counterexample without treating a reported Lean build as a premise. The repository's separate verification record distinguishes the source and build evidence for the external formalization from this independently reproduced arithmetic and exposition.